

IN GOLD THEY TRUST

GOLD PRICES, ARTISANAL MINING & SCHOOLING IN TANZANIA

ARTHUR RAINES

IN GOLD THEY TRUST

ABOUT THIS REPORT

This report is a revised version of my undergraduate thesis, *The Impact of Fluctuations in the Price of Gold on Education in Tanzania*. It uses the same data to answer the same question. However, following a review of the original analysis, it compares households within the same region and year. Section 8 describes each change and how it affects the results.

ABSTRACT

This paper evaluates the impact of fluctuations in the price of gold on school enrollment and household well-being in Tanzania. Using three rounds of the Tanzania National Panel Survey and a difference-in-differences strategy that compares households near and far from artisanal gold sites within the same region and year, I find that the ninety-one percent rise in the price of gold between 2008 and 2012 is associated with a 13.9 percentage point decline in enrollment among fourteen- to seventeen-year-olds living within thirty kilometers of an artisanal gold site. However, this estimate is not statistically significant. Its ninety-five percent confidence interval ranges from a decline of 34.1 percentage points to an increase of 6.3 percentage points. Additionally, the point estimates do not follow the path of the price. The gap appears in 2009, before the price had risen much, and it does not shrink in 2013, although these yearly estimates are imprecise. When I follow the same individuals over time, the estimate falls almost to zero. After I account for testing seven outcomes together, none of the outcomes I study remains statistically significant. In short, I do not find strong evidence that the gold boom reduced enrollment, although a large effect cannot be ruled out.

Keywords: Education; gold; artisanal mining; commodity prices; child labor; Tanzania

Abbreviations: ASM, artisanal and small-scale mining; IPIS, International Peace Information Service; NPS, Tanzania National Panel Survey; PML, Primary Mining Licence

HOW THE NUMBERS WERE MADE

I produced every estimate, table, and chart in this report with scripts written in Sift. The models are in *sift-revision/sift/revised.sift*, and the descriptive statistics are in *report/report.sift*. Before making any table, I checked each variable against the survey questions it came from (Appendix B). Gold prices are the World Gold Council's annual averages. The photographs are from Wikimedia Commons and are credited on page 30.

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AT A GLANCE

A boom in the price of gold

Between 2008 and 2012, the world price of gold rose ninety-one percent, from $872 to $1,669 an ounce, before falling fifteen percent in 2013. Over the same years, the Tanzania National Panel Survey interviewed the same households three times.

A smaller and imprecise effect on enrollment

When I compare households within the same region and year, I find that this rise in the price is associated with a 13.9 percentage point decline in enrollment among individuals fourteen to seventeen years old living within thirty kilometers of an artisanal gold site. However, this estimate is not statistically significant, and its ninety-five percent confidence interval ranges from a decline of 34.1 percentage points to an increase of 6.3 percentage points.

The timing of the gap

The gap in enrollment is already large in 2009, before the price had risen much, and it does not shrink in 2013, when the price fell. However, the estimates for these years are imprecise. When I follow the same individuals over time, the estimate falls almost to zero, although it is also imprecise.

Signs of an income effect

Households near the sites spent more as the price rose, and adults near the sites worked fewer hours on the farm. Enrollment among younger children also increased slightly. However, after I account for testing seven outcomes together, none of these estimates is statistically significant.

How this version differs from the original

This version revises my undergraduate thesis. It compares households within the same region and year, measures proximity at the 2008/09 location, clusters the standard errors by village, and measures the change in the price of gold in logs.



Separating gold-bearing soil near the Lupa River, Chunya district, 1951.

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THE PROMISE OF GOLD

Why might a gold boom change who goes to school?

Gold has been mined in Tanzania for more than a century, and hundreds of thousands of Tanzanians mine it by hand today. Between 2008 and 2012, the world price of gold rose by ninety-one percent. An increase in the price of gold may raise the opportunity cost of schooling for teenagers living near gold sites. However, it may also give their households more money to spend on school. In this part, I introduce the question, review what other studies have found, and describe the history of gold mining in Tanzania and the conditions in its artisanal mines today.

THE PROMISE OF GOLD
Introduction
Theoretical background
Gold in Tanzania

MEASURING THE BOOM

WHAT THE DATA SHOW

LOOKING AHEAD

1. INTRODUCTION

Gold's role and importance in Tanzania's economy grew rapidly in the years before this study. Between 2000 and 2008, minerals made up as much as forty-eight percent of the nation's merchandise exports, and gold production rose from less than one ton in 1998 to about fifty tons in 2008 (World Bank, 2009). Much of this gold comes from large industrial mines. However, hundreds of thousands of Tanzanians also mine gold by hand in small and often unlicensed pits. By 2011, there were an estimated 680,385 artisanal and small-scale miners in Tanzania, 27.6 percent of whom were women (Mutagwaba et al., 2018).

Between 2008 and 2012, the world price of gold rose by ninety-one percent. For a country like Tanzania, where much of its population works in agriculture, an increase in the price of gold may cause many to move away from agriculture toward mining. For a teenager living near a gold site, a higher price raises the opportunity cost of schooling, since a day spent in the pits pays more than before. However, higher prices also provide households with more money to spend, and households with more money may use the extra income to send their children to school. The consequences of such a shift on individual and household outcomes remain unclear. This question is especially important in a country that was rapidly expanding secondary schooling. The number of secondary school students in Tanzania rose from 524,325 in 2005 to 1,638,669 in 2010 (United Republic of Tanzania, 2010).

This paper aims to understand the consequences of fluctuations in the price of gold on schooling and household well-being in Tanzania. To do this, I compare individuals fourteen to seventeen years old whose households lived within thirty kilometers of a known artisanal gold site in 2008/09 with individuals of the same age in the same region who lived farther away. I use data on individual and household outcomes from three rounds of the Tanzania National Panel Survey, which interviewed the same households in 2008/09, 2010/11, and 2012/13. Over these years, the price of gold rose from $872 to $1,669 an ounce and then fell to $1,411 in 2013.

This paper revises the original version of my thesis. In the original version, I found that a sixty-two percent increase in the price of gold led to an 18.6 percentage point decline in enrollment among individuals fourteen to seventeen years old living within thirty kilometers of an artisanal gold mine. A review of that analysis raised several concerns. Most of the gold sites are in the Lake Zone, so the original model largely compared the Lake Zone with the rest of the country. Additionally, a household's proximity to a mine was measured at its location in each round, which changes when a household moves, and the standard errors were clustered at a level close to the household. In this version, I use the same data and question, but I compare households within the same region and year.

"Most of the gold sites are in the Lake Zone, so the original model largely compared the Lake Zone with the rest of the country."

I find that the ninety-one percent increase in the price of gold between 2008 and 2012 is associated with a 13.9 percentage point decline in enrollment among individuals fourteen to seventeen years old living within thirty kilometers of an artisanal gold site (see Table 2). However, this estimate is not statistically significant. Its ninety-five percent confidence interval ranges from a decline of 34.1 percentage points to an increase of 6.3 percentage points. Additionally, the point estimates do not follow the path of the price. The gap is already large in 2009, before the price had risen much, and it does not shrink in 2013, when the price fell by

fifteen percent. However, the estimates for these years are imprecise, and I cannot rule out that the gap follows the path the price implies. When I follow the same individuals from one round to the next, the estimate falls almost to zero, although it is also imprecise. I also find that household expenditure increases near artisanal sites as the price rises, and that adults near the sites work fewer hours on the farm. However, after I account for testing seven outcomes together, none of these estimates is statistically significant.

These findings contribute to the literature on natural resources and human capital in Africa (Ahlerup et al., 2020; Bazillier and Girard, 2018) by providing greater insight into the short run effects of changes in the price of a key export on schooling. They suggest that the regions where artisanal mining is concentrated were changing in other ways during these years, and that estimates which do not compare households within the same region may overstate the effect of the price of gold.

The rest of the paper is as follows. Section 2 discusses the literature and theoretical framework. Section 3 presents a history of gold mining in Tanzania and describes artisanal mining today. Section 4 describes the data. In Section 5, I outline the empirical methodology. Section 6 describes the main results of changes in the price of gold on enrollment. Section 7 investigates the mechanisms behind the results seen in Section 6. Section 8 describes how this version differs from the original and the limitations that remain. Section 9 concludes.

2. THEORETICAL BACKGROUND

Several empirical studies have focused on the impact of price fluctuations in commodities on human capital attainment. Cogneau and Jedwab (2012) find that a fall in cocoa prices in Côte d'Ivoire led to a decline in enrollment and an increase in child labor. Other studies suggest that human capital investment responds to shocks to agricultural production caused by natural events like droughts (Beegle et al., 2008; Yamano et al., 2005). In Tanzania, Beegle, Dehejia, and Gatti (2006) find that negative shocks to household income increase the number of hours children work.

The link between prices and human capital investment comes from several channels. An increase in the price of gold raises the return to working and, therefore, the cost of time spent in school. Meaning, household consumption of schooling falls. However, higher prices also



Small-scale miners rest inside a pit about one hundred meters deep at the Ihanzutwa gold mines in Iringa, 2014, with the journalist Nuzulack Dausen (front right), who reported on their work.

Photo: Ndausen / Wikimedia Commons, CC BY-SA 4.0

provide households with more money to spend. With more money to spend, households may use the extra income to send their children to school (Basu and Van, 1998; Baland and Robinson, 2000). Which effect dominates likely depends on who in the household earns the higher return. Meaning, if the price of gold mainly raises the earnings of teenagers, the substitution effect is likely to dominate, whereas if it mainly raises their parents' earnings, the income effect may dominate.

Gold mining and schooling in Africa

Recent work on gold mining in Africa finds evidence of both effects. Ahlerup, Baskaran, and Bigsten (2020) combine Afrobarometer surveys from thirty countries with the dates on which gold mines were discovered and closed. They find that individuals who had a gold mine in their district at the age of twelve have educational attainment 0.14 to 0.18 points lower, on a scale of zero to nine. The effect is largest at the secondary level and comes from smaller mines, which the authors attribute in large part to artisanal mining. They find no difference between men and women, and they suggest that child labor in mining districts is the main mechanism. Additionally, they find that the contemporaneous price of gold is unimportant for educational attainment.

Bazillier and Girard (2018) study the artisanal and industrial gold rushes in Burkina Faso. They find that a one percent increase in the price of gold leads to a 0.15 percent increase in the consumption of households living near artisanal mines, while industrial mines have no effect on the consumption of nearby households. However, they find no effect of the artisanal boom on the schooling of children up to sixteen years old, which I interpret as consistent with an income effect that offsets the pull of mining work. For individuals older than sixteen, they report a small decline in enrollment. They also find that mining acts as a seasonal complement to farming.

Recent work on human capital attainment from natural resource extraction also looks at the proximity of individuals, households, and communities to a mine on several measures of well-being (Hoedoafia et al., 2014; Kotsadam and Tolonen, 2016; Muslihudin et al., 2018; Zabsonré et al., 2017). For the most part, this literature is focused on the long-run consequences of living near a mine. This paper attempts to contribute to the literature by providing greater insight into the short run response of schooling to a large change in the price of gold, in a country where artisanal mining is widespread and legally recognized.

3. GOLD IN TANZANIA

3.1 A long history

Although it is rather hard to define artisanal and small-scale mining, artisanal and small-scale mining (ASM) generally involves using rudimentary tools like shovels, buckets, and pans on shallow ore deposits (Schwartz et al., 2021). It is estimated that about forty million people in eighty countries participate in ASM, and that ASM produces about twenty percent of the world's gold (Schwartz et al., 2021). In Tanzania, gold had an important yet limited role in the nation's early history. In the fourteenth and fifteenth centuries, the island of Kilwa Kisiwani, located in the southern half of the country, stood as the center of a gold trade in which gold from Zimbabwe was loaded and shipped to the Middle East and India (Pearson, 1998; Bryceson et al., 2012).

Gold production did not take place until 1898, when gold was discovered by a German company (Chachage, 1995), although the World Bank (2009) dates the first discoveries near Lake Victoria to around 1894. Production increased in the 1920s and 1930s due to a decline in agricultural prices and a rise in gold prices (Bryceson et al., 2012). Mining at this time was dominated by alluvial mining, which artisanal and small-scale miners primarily

perform (Roberts, 1986). Early estimates of the mining sector in 1938 found that of the 32,000 individuals working in the mining sector, seventy-three percent worked as small-scale miners (Lemelle, 1986). Several photographs in this report show small-scale mining on the Lupa goldfield, in the southwest of the country, in 1951.

The government in 1967 nationalized most of the country's mines (Bryceson et al., 2012). By then, most production came from artisanal mining (World Bank, 2009). The Mining Act of 1979 kept all minerals as the property of the state, but it gave more established rights to small-scale mining. Under the act, the Minister could set aside designated areas for mining "by methods not involving substantial expenditure or the use of specialist technology," in which citizens could hold a twelve-month prospecting right and register claims (United Republic of Tanzania, 1979). The government went further with the Small-Scale Mining Policy Paper of 1983, which encouraged individuals to participate in mining to supplement their incomes (Chachage, 1995). From 1987 to 1997, artisanal and small-scale miners produced almost all of the country's gold (Mutagwaba et al., 2018).

The Mineral Policy of 1997 and the Mining Act of 1998 changed the role of the government from owner and operator of the mines to regulator, and opened the sector to foreign investment (Mutagwaba et al., 2018). The 1998 Act created the Primary Mining Licence, a five-year license for areas of up to ten hectares, reserved for citizens (World Bank, 2009). However, as foreign companies took up exploration licenses in the Lake Victoria goldfields, miners without licenses were evicted (Mutagwaba et al., 2018). Large industrial mines opened, and national gold production rose from less than one ton in 1998 to about fifty tons in 2008 (World Bank, 2009). The Mineral Policy of 2009 and the Mining Act of 2010 again set out to develop small-scale mining, and the 2009 policy called for stronger enforcement of the laws on child labor in mining (Mutagwaba et al., 2018). In short, artisanal and small-scale mining in Tanzania, unlike in many other countries, is a legal and established alternative to agriculture.

3.2 Artisanal mining today

The number of artisanal and small-scale miners in Tanzania has grown steadily, from 150,000 in 1987 to 550,000 in 1996 and 680,385 in 2011 (Mutagwaba et al., 2018). About two-thirds of them mine gold, and nearly twenty-four percent



Panning (left) and washing gold-bearing sands in a sluice (right) on the Lupa River, Chunya district, 1951.

Photos: Touring Club Italiano archive / Wikimedia Commons, CC BY-SA 4.0

of all artisanal gold miners work in Geita Region alone (Mutagwaba et al., 2018). However, artisanal mining produces only about ten percent of the country's gold (Kinyondo and Huggins, 2020). The World Bank (2009) describes it as an activity that allows people to complement the income they earn from farming and other work.

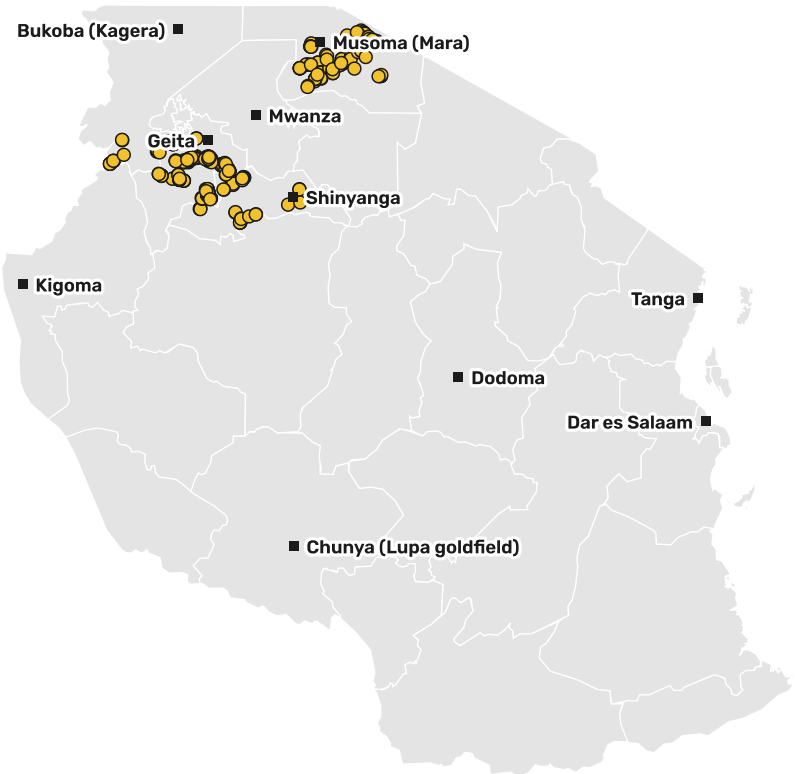
Most artisanal miners work outside the formal system. A Primary Mining Licence can cost up to a million shillings for ten hectares (Mutagwaba et al., 2018). By 2017, the government had designated thirty-six areas for artisanal mining, but only 626 of the licenses in these areas were active, and most license holders did not develop their areas themselves (Mutagwaba et al., 2018). For every miner who is formalized, an additional five informal miners are estimated to enter the sector (Mutagwaba et al., 2018). Additionally, the government's training "centres of excellence" for artisanal miners were built with World Bank support but were not functioning when the project ended, and the major legal reforms of 2017 targeted large-scale mining and ignored artisanal mining (Kinyondo and Huggins, 2020).

3.3 Health risks and child labor

Artisanal mining is a labor-intensive process with serious health risks and even death (Burki, 2019). Pits can reach seventy meters deep, and the sacks of ore that workers carry can weigh as much as sixty kilograms (O'Driscoll, 2017). Between 2008 and May 2017, Tanzanian authorities recorded 123 deaths at licensed sites and 140 deaths at unlicensed sites (Mutagwaba et al., 2018). More recently, in January 2024, the collapse of an unlicensed small-scale gold mine in Simiyu region after heavy rains killed at least twenty-two people (VOA, 2024).

Figure 1. Artisanal gold sites visited by IPIS

Each dot is one of the 337 gold sites IPIS visited in 2017 and 2018, over the district boundaries.



Gold sites cluster in the Lake Zone, south and east of Lake Victoria. The sites shown here are those in the IPIS data. The measure of proximity in this paper also uses small-scale gold mines recorded by the University of Glasgow, which were not available for this revision. These mines add households near a site in Tanga, Morogoro, Singida, and Manyara. Industrial mines recorded by Mindat enter separately, in column (9) of Table 2.

The Lupa goldfield near Chunya, where the photographs from 1951 were taken, lies in the southwest.

Sources: IPIS (2017–2018 site visits); district boundaries from the original version.

Mercury is widely used to separate gold from ore (World Bank, 2009). Miners use roughly half a tablespoon of mercury for each pan and burn it off to recover the gold (O'Driscoll, 2017). In the Rwamgasa area alone, an estimated twenty-seven kilograms of mercury are released into the environment each year, and mercury causes neurological damage to children (Mutagwaba et al., 2018). By 2013, artisanal gold mining had become the largest source of mercury released by people worldwide (Schwartz et al., 2021).

Children also work in the pits. Tanzanian law prohibits children under eighteen from working in mining (Employment and Labour Relations Act, 2004), and Tanzania ratified the International Labour Organization's conventions on the minimum working age and on the worst forms of child labor (Mutagwaba et al., 2018). Nevertheless, a 2014 survey found 30,827 children between five and seventeen years old working in artisanal and small-scale mining, and those fourteen to seventeen years old worked an average of twenty-eight hours a week (O'Driscoll, 2017). Mining accounts for 18.7 percent of the children in hazardous work in Tanzania, and inspections do not cover unlicensed gold sites (O'Driscoll, 2017). However, the link between mining and schooling is debated. Some researchers argue that income from mining often enables children to afford school (O'Driscoll, 2017).

INSIDE AN ARTISANAL GOLD SITE

In 2017 and 2018, researchers from the International Peace Information Service (IPIS) visited 337 artisanal gold sites in Geita, Mara, Shinyanga, and Kigoma regions. They counted 63,355 people working at these sites, nineteen percent of whom were women. Children under fifteen were working at 43 of the sites, 320 children in total. Mercury was used at 194 of the 237 sites where the question was answered, and 85 deaths from accidents were reported at 37 sites.

Source: IPIS open data, calculated for this report. The visits took place several years after the survey.



Boys washing gold-bearing sands from the Lupa River, 1951. Children have long worked in Tanzania's gold fields.

Photo: Touring Club Italiano archive / Wikimedia Commons, CC BY-SA 4.0



Students in a secondary school classroom, Kilimanjaro region. *Seemannaufland, CC BY-SA 4.0*

MEASURING THE BOOM

How can the effect of a world price be seen in Tanzanian villages?

The price of gold is the same for every household, so its effect can only be seen by comparing households that could and could not take advantage of it. I use three rounds of the Tanzania National Panel Survey, which interviewed the same households in 2008/09, 2010/11, and 2012/13, and I locate each household relative to known artisanal gold sites. In this part, I describe the survey and mine data, check each variable against the survey questions, and set out the model that compares households near and far from the sites within the same region and year.

THE PROMISE OF GOLD

MEASURING THE BOOM
Data
Methods

WHAT THE DATA SHOW

LOOKING AHEAD

4. DATA

4.1 Education data

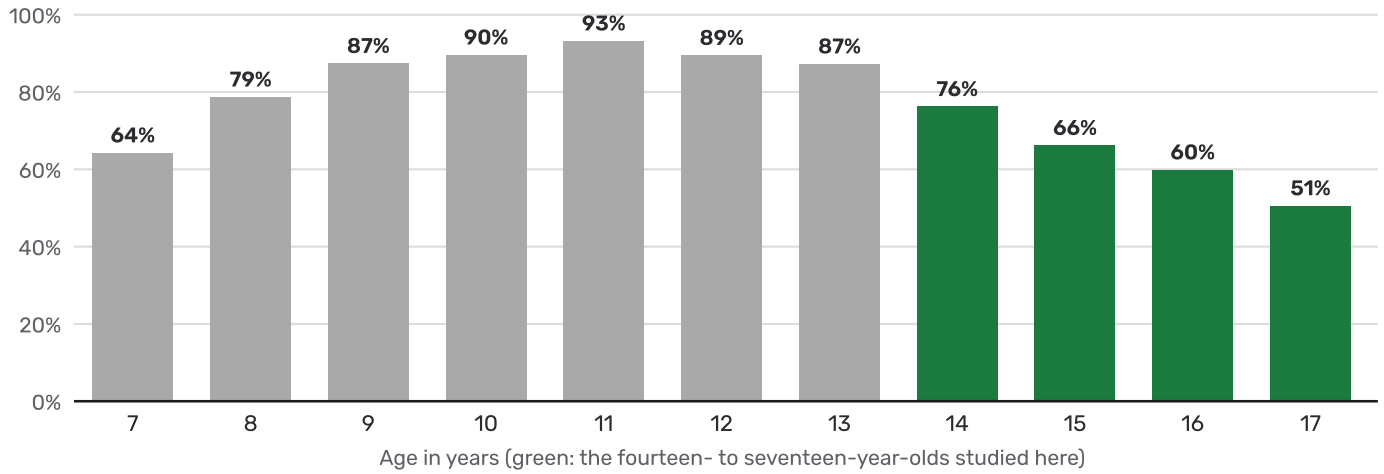
My analysis uses data from three rounds of the Tanzania National Panel Survey (NPS). The National Bureau of Statistics conducted all three rounds of the survey with technical support from the World Bank. The three rounds of the survey ran from 2008 to 2009, 2010 to 2011, and 2012 to 2013. Each survey round provides detailed information on household and individual characteristics, like whether an individual was enrolled in school at the time of the interview. The NPS also includes the location of each household interviewed in the survey. To ensure the anonymity of each household, its location is offset by up to two kilometers in urban areas and up to five kilometers in rural areas, with an additional offset of up to ten kilometers for one percent of rural clusters (National Bureau of Statistics, 2013).

I construct an unbalanced panel of 2,990 households interviewed in the first round. The original survey consisted of 3,265 households; however, households with no location or interview date were dropped from the panel. My main sample consists of the 4,555 observations of individuals fourteen to seventeen years old in these households across the three rounds.

My key measure of enrollment is constructed based on an individual's enrollment status at the time of the interview. Individuals that had no recorded response on their current enrollment status but responded to never attending school were recorded as not being enrolled. Individuals with missing responses on their current enrollment status but who responded to having attended school at some point in their life were dropped from the data. Of the 13,297 observations of individuals seven to seventeen years old across the three rounds, 10,429 reported being in school, 1,485 reported not being in school, and 1,383 had never attended school. Figure 2 plots the mean enrollment of individuals seven to seventeen years old. Individuals begin primary school at around age seven. Figure 2 shows that enrollment is highest from age ten to twelve, peaking at ninety-three percent at age eleven. From age fourteen to seventeen, there is a sharp decline in enrollment, from seventy-six percent to fifty-one percent.

Figure 2. Mean enrollment by age

Share of children in school at the time of the interview, all three rounds, weighted by the survey's household weights.



Source: Tanzania National Panel Survey 2008/09, 2010/11, 2012/13.

I measure household wealth as the log of per capita real expenditure, which is the household's total annual real expenditure in Tanzanian shillings divided by the number of people in the household. I measure labor force participation in each sector based on the total number of reported hours worked on the household farm and in a household business during the past seven days of the interview date. It should be noted that hours worked are only recorded in the first and third rounds, and they are missing for everyone in the 2010/11 round. Additionally, five responses report more than the 168 hours in a week, and I treat these responses as missing.

4.2 Mine data

Data on mine locations come from the International Peace Information Service (IPIS), Mindat, and the University of Glasgow. IPIS provides data on artisanal gold sites, the University of Glasgow provides data on small-scale gold mines, and Mindat provides data on industrial mines. As in the original version, mines that were discovered between 1920 and 1990 were dropped. These mines may have seen high usage during previous gold booms, and therefore there is likely to be little activity at them. Figure 1 plots the location of the IPIS gold sites.

A household is considered to be near an artisanal gold site if a known site lies within thirty kilometers of its location in 2008/09. I measure proximity at the first location so that a household that moves toward or away from the mines, possibly because of the price of gold itself, does not change its own proximity. Because thirty kilometers is large compared with the survey's location offsets, the offsets change the proximity of few households. Importantly, proximity to the sites is concentrated in a few regions. Only thirty-one of the three hundred and ninety-one survey villages with an individual fourteen to seventeen years old are within thirty kilometers of a site. Of the individuals fourteen to seventeen years old living near a site in 2008/09, seventy-six percent lived in Shinyanga, Mwanza, Kagera, and Mara, and every individual of that age surveyed in Mara lived near a site. For this reason, I compare households within the same region.

One caveat of the mine data is that the IPIS data does not represent sites observed within the survey years. The IPIS sites were recorded in 2017 and 2018, several years after the NPS. Therefore, I cannot say whether those sites were active during the survey years, and some may have opened because of the boom. I assume that the sites present potential mining opportunities within the area.

4.3 The price of gold

Data on gold prices come from the World Gold Council (see Figure 3). The annual average price of gold rose from $872 an ounce in 2008 to $972 in 2009, $1,225 in 2010, $1,572 in 2011, and $1,669 in 2012, an increase of ninety-one percent. In 2013, the price fell by fifteen percent to $1,411. Because each survey round ran over two calendar years, individuals in the same round were interviewed at different prices. I measure the price of gold as the log price in the interview year minus the log price in 2008. Meaning, the price measure is zero in 2008, 0.65 in 2012, and 0.48 in 2013.

4.4 Summary statistics

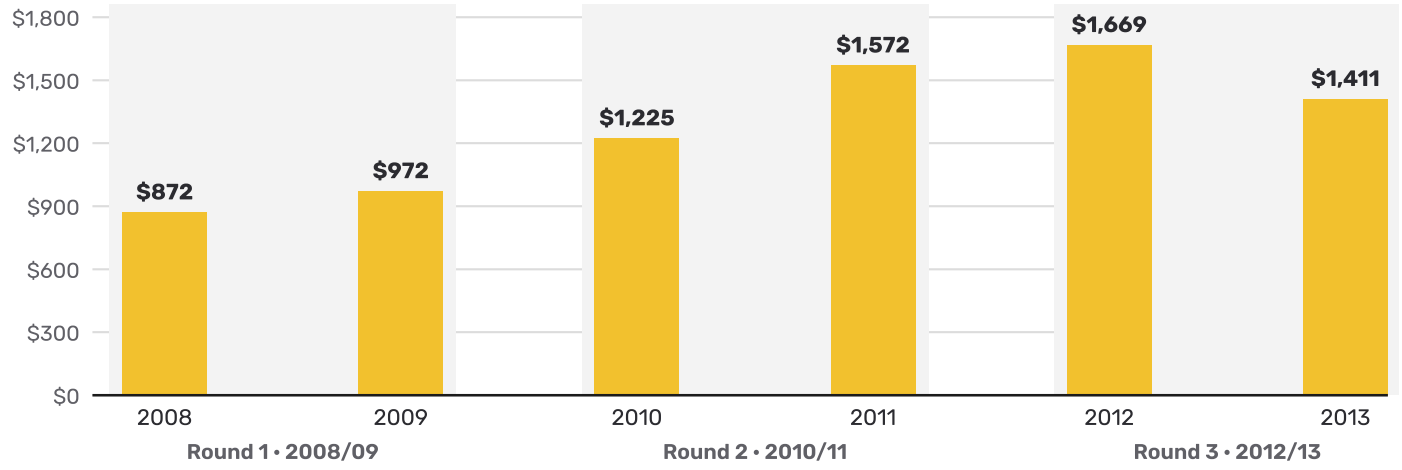
Table 1 provides summary statistics for individuals fourteen to seventeen years old across all three rounds. Of these individuals, 63.3 percent were enrolled at the time of the interview, and 15.4 percent lived within thirty kilometers of an artisanal gold site in 2008/09. All means are weighted using each round's household sampling weights. For every variable that also appears in Table 1 of the original version, the means are the same. The rows on proximity differ because proximity is now measured at the 2008/09 location, and the distance rows now describe separate rings

rather than overlapping distances. Appendix B describes the checks I made on each variable before building the table.

Figure 4 plots mean enrollment and mean log per capita real expenditure for those living within thirty kilometers of an artisanal gold site and those not. The figure shows that enrollment started off higher in 2008/09 for those living near a site (73.8 percent compared with 66.5 percent). By 2012/13, enrollment was lower for those living near a site (47.0 percent compared with 59.7 percent). Per capita expenditure rose for both groups. However, these comparisons do not account for other differences between the Lake Zone and the rest of the country. The model in Section 5 compares households within the same region and year.

Figure 3. Price of gold in each interview year

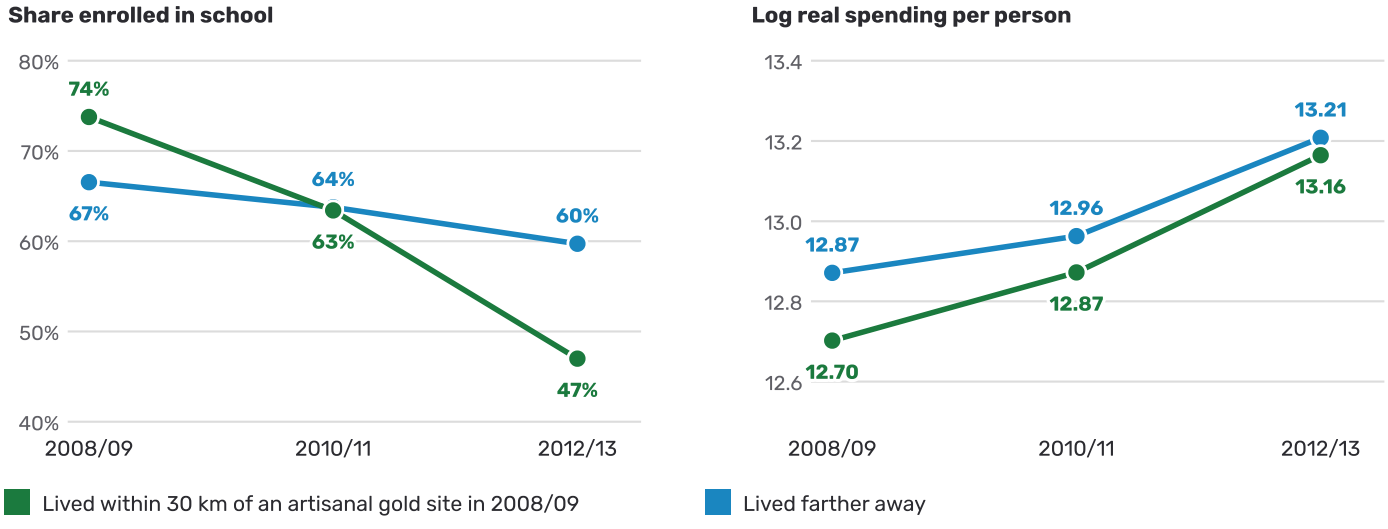
Annual average price, US dollars per troy ounce. Each survey round spans two interview years.



Source: World Gold Council annual averages, as recorded in the survey data.

Figure 4. Enrollment and spending of 14- to 17-year-olds, by survey round

Weighted means, by whether the household lived within 30 km of an artisanal gold site in 2008/09.



Source: Tanzania National Panel Survey; mine locations from IPIS, the University of Glasgow and Mindat.

5. METHODS

I begin the analysis by estimating the effect of fluctuations in the price of gold and proximity to an artisanal gold site on enrollment for individuals fourteen to seventeen years old. As seen in Figure 2, there is a large decline in enrollment from ages fourteen to seventeen. Individuals at this age must make critical decisions on whether to continue schooling or pursue other opportunities (Human Rights Watch, 2017). I estimate the following model below.

$$(1)\quad Enrollment_{iht} = \beta\,(Proximity_h \times P_t) + \delta_h + \delta_{rt} + \chi_{iht} + \varepsilon_{iht}$$

The dependent variable $Enrollment_{iht}$ is a dummy 1 if individual i in household h is enrolled in school in year t. $Proximity_h$ is a dummy 1 if a household's 2008/09 location is within thirty kilometers of a known artisanal gold site. P_t is the log price of gold in year t minus the log price in 2008. δ_h are household fixed effects, which account for every characteristic of a household that does not change over time, including its proximity to a site. δ_{rt} are region-by-year fixed effects, which account for changes that affect every household in a region in a given year, such as a regional drought. χ_{iht} is a set of controls that are unlikely to be affected by the price of gold, such as age

Table 1. Summary statistics, 14- to 17-year-olds

Variable	Obs.	Mean	SD
Enrolled in school	4,555	0.633	0.482
Artisanal gold site within 30 km (2008/09 home)	4,555	0.154	0.361
… within 0–10 km	4,555	0.058	0.233
… 10–20 km	4,555	0.041	0.198
… 20–30 km	4,555	0.055	0.228
… 30–40 km	4,555	0.027	0.162
… 40–50 km	4,555	0.046	0.209
… 50–60 km	4,555	0.048	0.213
Industrial gold mine within 30 km	4,555	0.077	0.267
Age (years)	4,555	15.485	1.112
Female	4,555	0.497	0.500
Rural	4,555	0.775	0.417
Annual rainfall (mm)	4,555	787.79	228.95
Annual mean temperature (°C × 10)	4,555	222.42	25.81
Household head married	4,555	0.633	0.482
Household head works in agriculture	4,555	0.715	0.451
Household moved	4,555	0.027	0.161
Log real spending per person (TSh)	4,553	12.991	0.656
Farm hours, past seven days	3,051	8.213	14.551
Non-farm business hours, past seven days	3,046	7.583	12.412

Notes: Means and standard deviations are weighted with each round's household sampling weights. Distances are from the household's 2008/09 location. Hours are recorded in 2008/09 and 2012/13 only. Sources: Tanzania National Panel Survey; IPIS; Mindat; University of Glasgow.

and gender dummies, the interaction between age and gender, and the annual mean precipitation and annual mean temperature. Lastly, ε_{iht} is the error term.

The coefficient of interest, β, measures the change in enrollment for those living within thirty kilometers of an artisanal gold site versus those not, for a one-unit increase in the log price of gold, compared with other households in the same region. Multiplying β by 0.65 gives the change in enrollment from 2008 to 2012. The regression is weighted using the survey's household sampling weights, and the standard errors are clustered at the village level, since proximity is assigned by village. Households observed only once in the sample are dropped, because their household fixed effect fits them exactly.

Unlike the original version, the model does not control for whether a household moved, whether the household head works in agriculture, the marital status of the household head, or whether a household lives in an urban or rural area. An increase in the price of gold may change each of these. For example, if adults move from farming into mining, as the original version suggested, then controlling for whether the household head works in agriculture would absorb part of the effect of the price of gold.

The timing of the effect

The key assumption of this model is that, without the changes in the price of gold, enrollment for those living near and far from the sites would have followed the same path within each region. I cannot test this before 2008, because the survey begins that year. However, the path of the price within the survey years provides a test. The price had risen little by 2009, and it fell in 2013. If the gap in enrollment between those near and far from the sites is caused by the price, it should be small in 2009 and should shrink in 2013. Therefore, I also estimate an event study, in which proximity is interacted with each interview year.

$$(2)\quad Enrollment_{iht} = \Sigma_s \theta_s \, (Proximity_h \times Year_{st}) + \delta_h + \delta_{rt} + \chi_{iht} + \varepsilon_{iht}$$

$Year_{st}$ is a dummy 1 if the interview took place in year s, for each year from 2009 to 2013. Each θ_s measures the gap in enrollment in year s compared with 2008. If the gap is caused only by the price of gold, θ_s should equal β times the log price in that year.



Makeshift sails send fresh air down into the pits at an artisanal gold mine near Dodoma, 1997. *Hans Erren, CC BY-SA 4.0*

WHAT THE DATA SHOW

Did teenagers near the gold sites leave school as the price rose?

The original version of this paper found a large decline in enrollment near artisanal gold sites. In this part, I move from the original model to the preferred model one step at a time, and I test whether the gap in enrollment follows the price of gold over time and fades with distance from a site. I then look at the mechanisms through which the price could affect schooling, such as household expenditure, the hours adults and teenagers work, and the enrollment of younger children.

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LOOKING AHEAD

6. RESULTS

6.1 The impact of gold price on enrollment

Table 2 shows the impact of gold price on enrollment. Each column changes one part of the model, moving from the original version to equation (1). Column (1) represents the original model. It gives the same coefficient as the original version, although the standard error and the number of observations differ slightly because households observed only once are dropped. It suggests that a one-unit increase in the log price of gold leads to a −0.339 change in enrollment for those living within thirty kilometers of an artisanal mine versus those not. In columns (2) to (4), I measure proximity at the 2008/09 location, cluster the standard errors by village, and remove the controls that the price of gold may affect. These changes have little effect on the estimate, which is −0.380 in column (4). Figure 7 in Appendix A plots the estimate from each column.

In column (5), I add region-by-year fixed effects, which gives equation (1). The estimate falls by forty-four percent to −0.214, with a standard error of 0.158, and it is no longer statistically significant (p = 0.176). This result suggests that much of the original estimate came from other changes in the regions where the gold sites are located. During the survey period, the log price of gold rose by 0.65 between 2008 and 2012. Therefore, the estimate in column (5) implies a 13.9 percentage point decline in enrollment among individuals fourteen to seventeen years old living within thirty kilometers of an artisanal gold site. However, the ninety-five percent confidence interval ranges from a decline of 34.1 percentage points to an increase of 6.3 percentage points.

Columns (6), (7), and (9) test the preferred estimate. In column (6), clustering the standard errors by district instead of village gives a smaller standard error of 0.126 and a p-value of 0.092. In column (7), removing the sampling weights has little effect on the estimate. In column (9), when controlling for proximity to an industrial mine, the estimate for artisanal sites is −0.186 and the estimate for industrial mines is −0.088. Neither estimate is statistically significant.

Column (8) follows the same individuals from one round to the next by replacing household fixed effects with individual fixed effects. The estimate falls to −0.024, with a standard error of 0.176. With household fixed effects, the model compares a fifteen-year-old in one round with a younger sibling who turns fifteen in a later round. This comparison mixes the effect of the price of gold with changes in which children in a household fall within the age group. Meaning, when I follow the same individuals, the estimate falls almost to zero. However, the sample is smaller (2,885 observations), and the estimate is imprecise.

6.2 The timing of the effect

Figure 5 shows the results of the event study in equation (2). If the gap in enrollment were caused by the price of gold, it would be small in 2009, when the price had barely risen, and it would shrink in 2013, when the price fell. However, the gap is already −0.170 in 2009, while the price alone would imply a gap of −0.023. The gap is −0.248 in 2010, −0.218 in 2011, and −0.338 in 2012, and it remains at −0.323 in 2013, while the price alone would imply −0.103. Only the estimate for 2012 is statistically significant at the five percent level. This finding suggests that the gap may reflect different trends for those living near and far from the sites. However, because the estimates for 2009 and 2013 are imprecise, I cannot rule out that the gap follows the path the price implies.

6.3 Distance

Figure 6 shows the impact of price fluctuations at various distances from artisanal gold sites. If high gold prices draw teenagers out of school

near the mines, the effect should be strongest close to a site and fade at greater distances. Unlike the original version, which estimated a separate regression for each distance, I estimate all six distances in one regression. Meaning, each distance is compared with households more than sixty kilometers from a site rather than with households at the other distances. The estimates are 0.070 within ten kilometers, −0.158 at ten to twenty kilometers, −0.352 at twenty to thirty kilometers, 0.421 at thirty to forty kilometers, −0.082 at forty to fifty kilometers, and 0.051 at fifty to sixty kilometers. This finding suggests that the effect does not fade with distance. The only statistically significant estimate, at thirty to forty kilometers (p = 0.041), is positive. As in the original version, the reason for this increase in enrollment at these distances remains unclear.

7. MECHANISMS

The findings in Table 2 show that, once households are compared within the same region and year, the decline in enrollment is smaller and not statistically significant. This section tests the channels through which the price of gold could still affect households near the sites. I begin by looking at household wealth for households with individuals fourteen to seventeen years old. From there, I look at the hours worked by adults and by individuals fourteen to seventeen years old, and at the enrollment of younger children. Table 3 reports the results for every outcome.

Table 2. Impact of gold price on enrollment, 14- to 17-year-olds, one change at a time

Model	β	SE	p	Obs.
(1) The original model	−0.339***	(0.123)	0.006	4,141
Current exposure, all controls, village-round clusters, year effects				
(2) Exposure fixed at the 2008/09 home	−0.347***	(0.123)	0.005	4,141
Price measured from 2008				
(3) Clustered by village	−0.347***	(0.132)	0.009	4,141
Village of the 2008/09 home				
(4) Only controls the price cannot move	−0.380***	(0.133)	0.005	4,141
Weather, age by sex				
(5) Region-by-year effects	**−0.214**	**(0.158)**	**0.176**	**4,136**
Preferred model, equation (1)				
(6) As (5), clustered by district	−0.214*	(0.126)	0.092	4,136
(7) As (5), without sampling weights	−0.219	(0.152)	0.151	4,136
(8) As (5), the same child followed	−0.024	(0.176)	0.893	2,885
Person instead of household effects				
(9) As (5), with industrial mines	−0.186	(0.176)	0.290	4,136
In (9), industrial gold mine within 30 km × log gold price: −0.088 (0.205), p = 0.670				

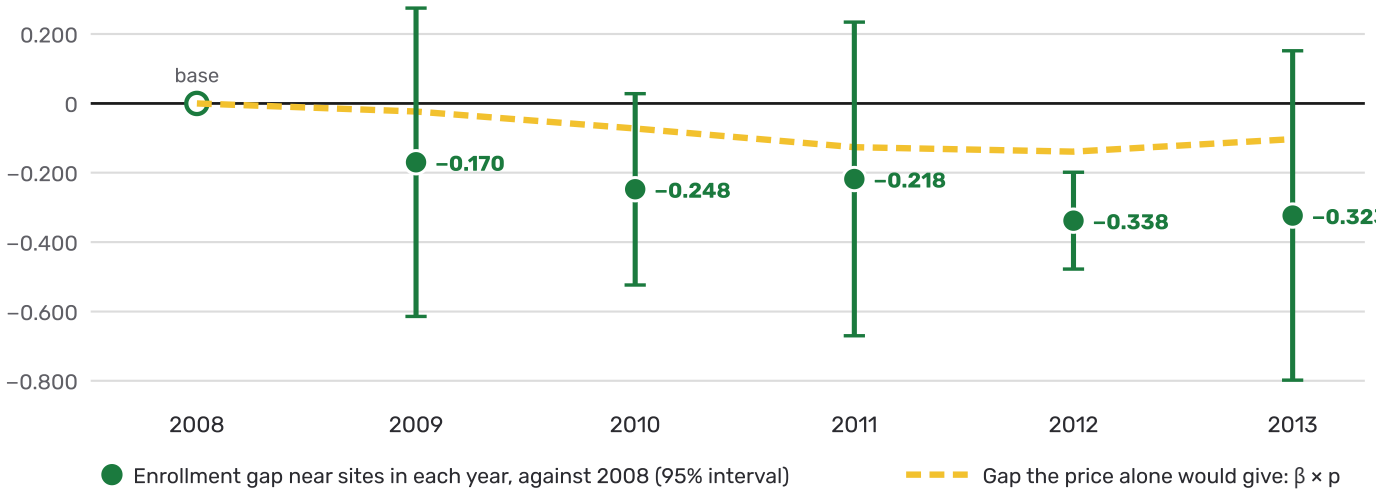
*Notes: β is the coefficient on proximity to an artisanal gold site (within 30 km) times the log price of gold. Every column has household fixed effects (column 8: individual fixed effects), age-by-sex dummies, rainfall and temperature. Columns (1) to (4) have interview-year effects; columns (5) to (9) have region-by-year effects. Columns (1) and (2) cluster by village and round; columns (3) to (9) by village, except column (6), by district. Sampling weights except column (7). Standard errors in parentheses. Column (1) drops households observed only once, so its standard error and number of observations differ slightly from the original version (0.129 and 4,555). *significant at 10% **significant at 5% ***significant at 1%. Sources: Tanzania NPS, IPIS, Mindat, University of Glasgow, World Gold Council.*

7.1 The impact of gold price on per capita real expenditure

High gold prices may improve household well-being through higher wages. Higher wages raise the opportunity cost of schooling. If the substitution effect dominates, enrollment declines. However, if the income effect dominates, enrollment will increase. Table 3 tests whether household wealth, measured as log per capita real expenditure, increases. The sample contains only households with individuals fourteen to seventeen years old. The results suggest that a one-unit increase in the log price of gold leads to a 0.266 increase in log per capita real expenditure for households within thirty kilometers of an artisanal site versus those not (p = 0.120). Between 2008 and

Figure 5. The enrollment gap near gold sites, year by year

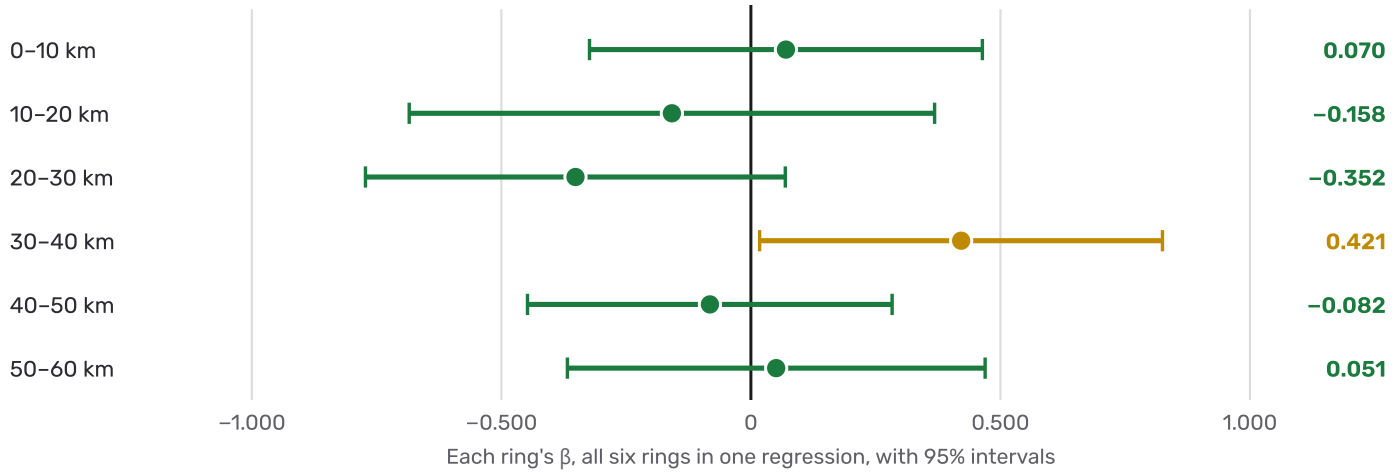
Equation (2): the gap in each interview year against 2008, with 95% confidence intervals, and the gap the gold price alone would produce.



Notes: 14- to 17-year-olds; controls as in equation (1). Sources: Tanzania NPS, IPIS, University of Glasgow, World Gold Council.

Figure 6. The impact of gold price on enrollment by distance to an artisanal gold site

One regression with every ring's proximity times the log gold price; 14- to 17-year-olds; 95% confidence intervals.



Notes: as Figure 5. Sources: Tanzania NPS, IPIS, Mindat, University of Glasgow, World Gold Council.

2012, this implies that per capita expenditure rose about nineteen percent more for households near the sites. However, the estimate is not statistically significant.

7.2 The impact of gold price on hours worked

Though individuals fourteen to seventeen years old may consume less education, it is unclear if this is due to substitution towards mining or other activities like agriculture. For adults eighteen to fifty-five years old, a one-unit increase in the log price of gold leads to a change of −10.22 hours of farm work a week for those living within thirty kilometers of an artisanal site versus those not (p = 0.059). Between 2008 and 2012, this implies about 6.6 fewer hours of farm work a week. This finding suggests that high gold prices may induce adults near artisanal sites to substitute away from agriculture for other activities like mining. For individuals fourteen to seventeen years old, the estimate for farm hours is positive (7.71, p = 0.318). This could come from teenagers filling in on the farm as their parents work more outside the home. Non-farm hours change little for either group. It should be noted that hours are only recorded in the 2008/09 and 2012/13 rounds, and none of these estimates is statistically significant at the five percent level.

7.3 Younger children

Although the gains in household expenditure do not lead to higher enrollment among individuals fourteen to seventeen years old, enrollment among children eight to twelve years old increases slightly near the sites (0.080, p = 0.275). Between 2008 and 2012, this implies an increase of about 5.2 percentage points. This finding would suggest that households invest more in younger children, which is consistent with much of the literature on household human capital investment (Akresh et al., 2012). However, the estimate is not statistically significant.

7.4 Testing several outcomes together

Because I test seven outcomes, one of them may appear statistically significant by chance. Table 4 and Figure 8 in Appendix A adjust the p-values for testing seven outcomes together. Holm's method controls the probability of any false finding, and the Benjamini–Hochberg method controls the expected share of false findings. The smallest unadjusted p-value, for adult farm hours, is 0.059. After the adjustment, no p-value falls below 0.41. In short, after the adjustment, none of the outcomes in Table 3 shows a statistically significant response to the price of gold.

Table 3. Impact of gold price on each outcome, equation (1)

Outcome	Ages	β	SE	p	Obs.	Change, 2008–2012
Enrolled in school	14–17	−0.214	(0.158)	0.176	4,136	−13.9 percentage points
Log real spending per person	Households of 14–17	0.266	(0.171)	0.120	4,134	+0.173 log points
Enrolled in school	8–12	0.080	(0.074)	0.275	5,867	+5.2 percentage points
Farm hours, past seven days	18–55	−10.220*	(5.400)	0.059	12,942	−6.6 hours a week
Non-farm business hours	18–55	−1.924	(2.509)	0.443	12,921	−1.2 hours a week
Farm hours, past seven days	14–17	7.708	(7.700)	0.318	2,324	+5.0 hours a week
Non-farm business hours	14–17	−1.591	(3.551)	0.654	2,318	−1.0 hours a week

*Notes: Each row estimates equation (1) for one outcome. The last column multiplies β by the 2008–2012 change in the log price of gold (0.65). Hours are recorded in 2008/09 and 2012/13 only. *significant at 10% **significant at 5% ***significant at 1%.*



Men queue at the permit office for the controlled area of the Lupa goldfield, 1951.

Touring Club Italiano archive, CC BY-SA 4.0

LOOKING AHEAD

What do the results mean, and what would settle the question?

The results of this version differ from those of the original version of this paper. In this part, I describe each change from the original version and how it affects the estimate, and I describe the limitations that remain. I then conclude with what the findings suggest for policy toward artisanal mining in Tanzania and for future research.

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8. REVISIONS AND LIMITATIONS

This section describes how this version of the paper differs from the original and why the estimates differ. I begin by rerunning the original analysis. It gives the same coefficient as the original version, −0.339. However, the standard error is 0.123 rather than 0.129, and the sample has 4,141 observations rather than 4,555, because households observed only once are dropped. The remaining differences come from several changes to the model.

The largest change in the estimate comes from comparing households within the same region. Because the artisanal gold sites are concentrated in a few regions, the original model with year fixed effects largely compared the Lake Zone with the rest of the country. In this version, region-by-year fixed effects compare households only with other households in the same region and year. As Table 2 shows, this change accounts for most of the difference between the two estimates.

Additionally, I measure proximity at a household's 2008/09 location. In the original version, proximity was measured at a household's location in each round. Because of this, proximity changed when a household moved or when the survey assigned a new location offset. Since the original model included household fixed effects, these changes alone identified the original estimate for proximity to an artisanal mine. I also remove the controls that the price of gold may affect, such as whether a household moved, the occupation and marital status of the household head, and whether a household lives in an urban area.

The original version clustered the standard errors at the village and year level. However, the village variable in the original data took a different value for nearly every household, so each cluster was close to a single household in a single round. In this version, I cluster the standard errors by the village of the 2008/09 location. Lastly, the original version multiplied the estimate by sixty-two, the percentage increase in the price of gold, to calculate the change in enrollment. Because the price is measured in logs, the estimate should instead be multiplied by the change in the log price. I also estimate the distance rings in one regression, add individual fixed effects and an event study, adjust the p-values for testing several outcomes, and treat impossible hours as missing.

Limitations

Several limitations remain. First, the IPIS sites were recorded in 2017 and 2018, after the survey. Some sites may have opened because of the boom, which would make proximity partly a result of the price of gold. Second, only thirty-one villages are near a site. With so few villages near a site, clustered standard errors may be too small, and a wild cluster bootstrap would provide a better test. Third, because the price of gold is the same for every household, any other change that affected the gold regions during the same years could still bias the estimate. Fourth, hours worked are missing for the 2010/11 round, and individuals who had attended school but did not report their current enrollment were dropped from the data. Lastly, the original version's analysis of expenditure for all households used a household-level data file that was not available for this revision, so this version studies expenditure only for households with individuals fourteen to seventeen years old.

9. CONCLUSION

This paper analyzed the impact of fluctuations in the price of gold and proximity to an artisanal gold site on education and household outcomes in Tanzania. My results show that, when households are compared within the same region and year, an increase in the price of gold is associated with a decline in enrollment among individuals fourteen to seventeen years old

living near an artisanal gold site. However, the estimate is smaller than in the original version and is not statistically significant. Additionally, the gap in enrollment appears in 2009, before the price of gold had risen much, and it does not shrink in 2013, although these yearly estimates are imprecise. When I follow the same individuals over time, the estimate falls almost to zero. A large effect remains possible, since the confidence interval includes a decline of 34.1 percentage points. Importantly, these findings show only the short-run consequences of increasing gold prices.

"If the government wishes to empower small-scale miners, it must lower entry barriers into the formal marketplace."

From a policy standpoint, my findings provide some insight. Households near artisanal sites spent more as the price of gold rose, and adults near the sites worked fewer hours on the farm. This suggests that small-scale mining could allow households to diversify their income stream. With that being said, most artisanal miners work outside the formal system. Licenses are costly, most license holders do not develop their areas, and the training centers meant to support small-scale miners were not functioning when the World Bank project that funded them ended (Mutagwaba et al., 2018; Kinyondo and Huggins, 2020). Additionally, child labor inspections do not cover unlicensed gold sites (O'Driscoll, 2017). If the government wishes to empower small-scale miners and protect the children who live near the mines, it must lower entry barriers into the formal marketplace and extend inspections to unlicensed sites.

Future research should investigate these questions with better data. Records of when artisanal sites opened would show whether proximity to a site itself responded to the price of gold. The fourth round of the survey, in 2014/15, when the price of gold continued to fall, would provide a stronger test of whether enrollment follows the price. Lastly, there may be persistent long-run consequences to individual, household, and community well-being (Ahlerup et al., 2020). Future research should investigate these long-run effects.



Maize fields between Kahama and Nzega in Shinyanga region. Most of the teenagers in the survey live in households whose head works in agriculture, and artisanal mining lets such households add a second source of income (World Bank, 2009).

Photo: Mamatimo / Wikimedia Commons, CC BY-SA 4.0

A. TABLES

Table 4 lists the adjusted p-values discussed in Section 7.4, and Figure 8 plots them. Figure 7 plots the estimate from each column of Table 2. Tables 1 to 3 appear in the text.

B. DATA CHECKS

Before using the analysis file from the original version, master_cleaned.dta, I checked each variable against the survey questions it came from. The checks are in *report/audit.sift*, and they report only counts.

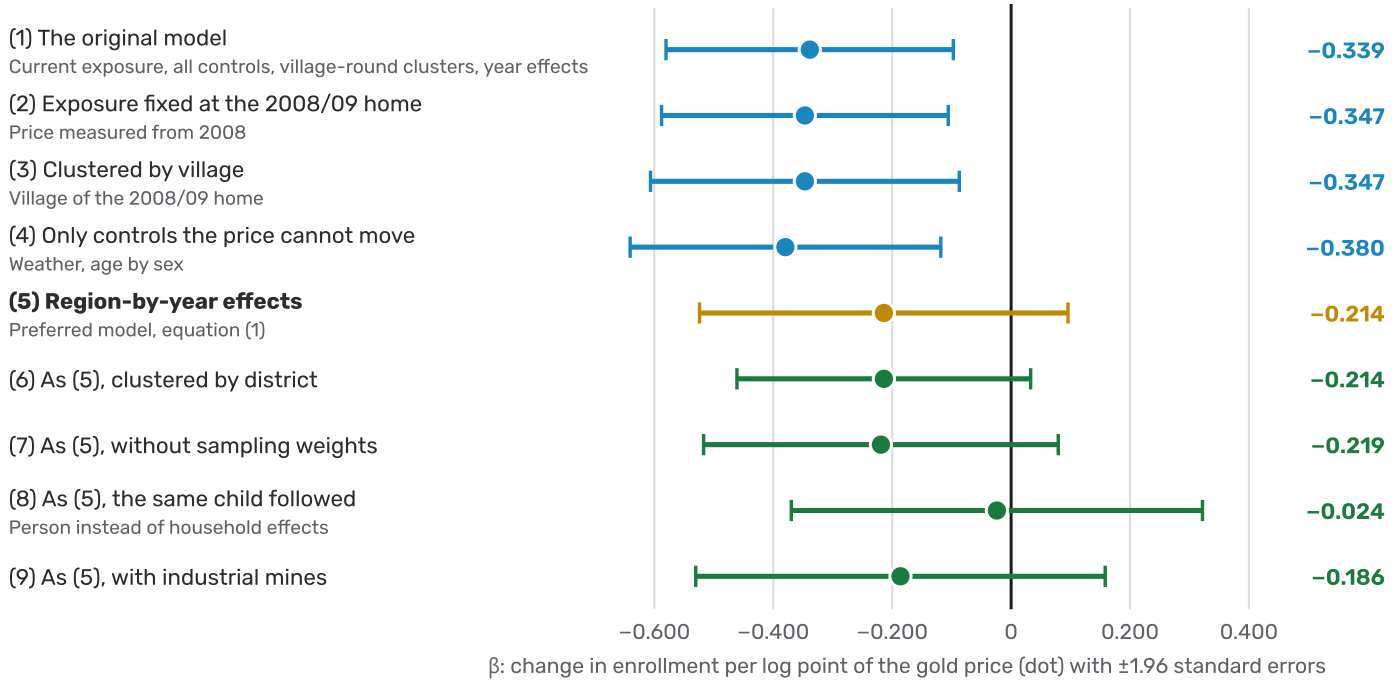
Table 4. p-values of β in Table 3, adjusted for seven tests

Outcome	p	Holm	Benjamini–Hochberg
Farm hours, 18–55	0.059	0.414	0.412
Log expenditure, 14–17	0.120	0.721	0.412
Enrollment, 14–17	0.176	0.882	0.412
Enrollment, 8–12	0.275	1.000	0.445
Farm hours, 14–17	0.318	1.000	0.445
Non–farm hours, 18–55	0.443	1.000	0.517
Non–farm hours, 14–17	0.654	1.000	0.654

Notes: Holm's step-down method controls the probability of any false finding among the seven; the Benjamini–Hochberg method controls the expected share of false findings. Outcomes are listed from the smallest unadjusted p-value.

Figure 7. The estimate as the model changes

Change in enrollment per log point of the gold price, near artisanal gold sites against farther away. Blue: the original model and columns (2) to (4); gold: the preferred model, column (5); green: columns (6) to (9).



Notes: Each row is one regression of enrollment of 14- to 17-year-olds; Table 2 gives the numbers. Sources: Tanzania NPS, IPIS, Mindat, University of Glasgow, World Gold Council.

First, no individual appears more than once in the same round, and the female dummy matches the survey's question on sex in every row. Second, the enrollment measure is coded as described in Section 4.1. Of the 13,297 observations of individuals seven to seventeen years old across the three rounds, every individual coded as enrolled reported being in school (10,429), and every individual coded as not enrolled either reported not being in school (1,485) or had never attended school (1,383). Third, in every row, age equals the interview year minus the reported year of birth, and an individual's age rises by one to three years between rounds, about two years on average.

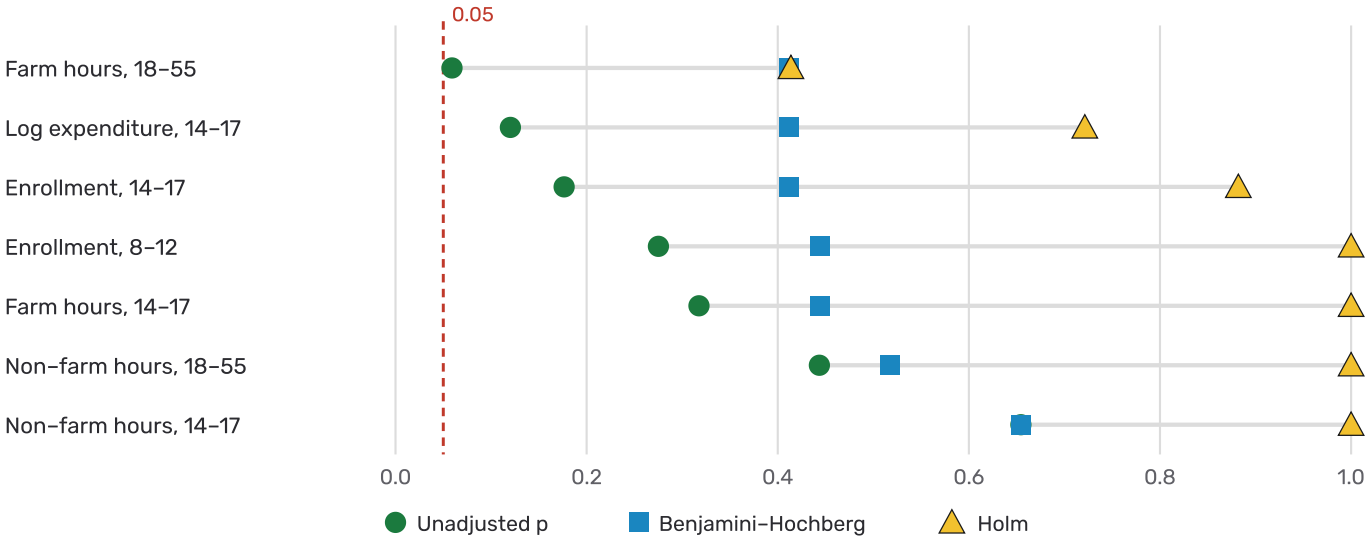
Additionally, wherever both are recorded, log per capita real expenditure equals the log of household real expenditure divided by household size. In the 2010/11 round, the moved dummy equals one exactly when the survey records a positive distance from the 2008/09 location. The household weights are constant within a household and round, but they are each round's own weights rather than the 2008/09 weights.

However, the hours data need attention. Farm and non-farm hours are missing for everyone in the 2010/11 round, including 6,561 adults eighteen to fifty-five years old and all individuals fourteen to seventeen years old. Additionally, five responses report more than 168 hours in a week, the largest being 999. I treat these responses as missing.

In short, the weighted means in Table 1 match those in Table 1 of the original version for every variable that both tables report, so the original summary statistics were computed correctly from this file. The original Table 1 also listed overlapping distances (within ten, twenty, and up to sixty kilometers) and a "monopsony mine" variable that was not defined in the text. Table 1 in this version reports separate distance rings and leaves out the undefined variable.

Figure 8. p-values before and after adjusting for seven tests

Each outcome's p-value from Table 3, and the same p-value adjusted with Holm's and the Benjamini–Hochberg methods.



Notes: Table 4 gives the numbers.

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Lupa goldfield, 1951 (pages 6, 9, 24 and the back cover): "Blowing selection of auriferous soils near the Lupa River" (BBU 097); "A man with a metal tool for gold sand sorting on the Lupa River" (BBU 092); "A man washing auriferous sands on the Lupa River" (BBU 094); "Indigenous boys engaged in the continuous washing of auriferous sands from the Lupa River" (BBU 096); "Men queuing at the permit office for residence in the controlled area of Lupa" (BBU 117); "A miner from New Saza Mines at work drilling quartz rock" (BBU 116). Unknown photographer, Archivio Storico del Touring Club Italiano. CC BY-SA 4.0.

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This paper evaluates whether the rise in the price of gold between 2008 and 2012 reduced school enrollment among teenagers living near artisanal gold sites in Tanzania. Using three rounds of the Tanzania National Panel Survey, I compare households near and far from the sites within the same region and year. I find a smaller decline in enrollment than in the original version of this paper, and the estimate is not statistically significant.

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